

**FLOODSENSE: AN ANDROID-BASED EXPERT SYSTEM FOR FLOOD
SUSCEPTIBILITY ASSESSMENT AND PRE-EVENT PREPAREDNESS IN
BACOR CITY**

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**REYMART V. GOC-ONG
MARTIN LORENZ D. JUANITES
ROMAR T. PULAO**
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**FloodSense: An Android-Based Expert System for Flood Susceptibility
Assessment and Pre-Event Preparedness in Bacoor City**

**Reymart V. Goc-ong
Martin Lorenz D. Juanites
Romar T. Pulao**

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INTRODUCTION

Flooding is one of the most frequent and destructive natural hazards in the Philippines. The country's location within the typhoon belt exposes communities to rainfall events that cause inundation, displacement, damage to property, interruption of livelihood, and loss of life. National disaster statistics continue to document the widespread effects of natural hazards on Filipino families and communities (Philippine Statistics Authority [PSA], 2026). Bacoor City, Cavite, is among the localities that regularly experience flooding. Its low-lying areas and connection to the Imus and Bacoor river systems contribute to persistent inundation during periods of heavy rainfall. Flood-mitigation facilities in the Imus River Basin were established specifically to reduce peak floodwater volume in Cavite, including low portions of Bacoor and its tributaries (Japan International Cooperation Agency [JICA], 2021). These conditions establish the importance of accessible, understandable, and location-specific information that residents can consult before a flood event develops.

To assess the information problem, the researchers surveyed 50 residents across 12 barangays in Bacoor City, with respondents drawn largely from historically flood-affected communities. The findings revealed that flood warnings are commonly disseminated through Short Message Service (SMS), social media, and barangay public-address systems, yet these channels do not consistently provide location-specific susceptibility information before flooding occurs. Only 20% of respondents felt well-informed about which areas in the city are flood-prone, meaning that 80% selected Somewhat Aware, Not Sure, or Not Aware at All. Most respondents also rated the timeliness, clarity, and quality of available flood information near the dissatisfied end of the scale. Most critically, 64% reported making a wrong decision, such as taking a flooded road, delaying evacuation, or misjudging flood conditions, because adequate area-specific information was unavailable. A key informant interview with the Bacoor Disaster Risk Reduction and Management Office (BDRRMO) confirmed that historical flood records and Geographic Information System (GIS)-based exposure information

are maintained institutionally but are not presented to residents through an interactive public platform. A supplementary interview with the City Planning and Development Coordinator Office (CPDCO) also established the availability of elevation, flood hazard, land-use, and barangay-boundary information relevant to geographic assessment.

FloodSense addresses this gap through an Android application that connects expert-provided and expert-validated knowledge with the information and preparedness needs of Bacoar City residents. A Rule-Based Expert System organizes approved facts and conditional rules in a knowledge base and applies an inference process to the user's selected rainfall intensity, rainfall duration, and geographic area. Rule-based expert systems use a knowledge base and an inference procedure that applies rules to known facts to reach a conclusion (Buchanan & Shortliffe, 1984). In FloodSense, the conclusion is an explainable flood susceptibility assessment expressed as Low, Moderate, High, or Very High. The Decision Support System (DSS) then uses the assessment together with the user's situational inputs to present suitable pre-event preparedness guidance. This function supports, but does not make, the user's final decision; decision support systems organize relevant information and structured guidance for human decision-making (Power, 2002).

The current FloodSense prototype presents scenario-based susceptibility assessment through an interactive map. The user selects or confirms a hypothetical rainfall intensity and duration, selects a supported location manually or through an optional one-time foreground location request, and explicitly requests an assessment. The map can display demonstration susceptibility zones and a neutral reference layer representing the 47 current barangay identities; the reference layer is technically validated for geometry but is not City-issued, is not City-verified, and contains no flood-susceptibility facts. The custom Web Administration System currently provides authenticated dashboard access, read-only review of map and rainfall-reference data, parameter-governance views, and controlled workflows for Decision Support System content, evacuation-center records, source records, and audit history. Ordinary

administrators do not edit raw Expert System rules, priorities, conflict-resolution logic, or algorithm code. A fixed-schema CSV export-and-import workflow for permitted values is a confirmed requirement, but it is not yet implemented. FloodSense does not provide real-time monitoring, official forecasting, autonomous alerts, or automatic evacuation orders.

Statement of the Problem

This study aimed to develop an Android application that provides location-specific flood susceptibility assessment and pre-event preparedness support for Bacoor City through a Rule-Based Expert System, a Decision Support System, and a dynamic geographic map. The research questions were grounded in the survey of 50 residents across 12 barangays, key informant interviews with the BDRRMO and CPDCO, and a review of related literature and official information. These sources were used to establish community needs, institutional knowledge sources, geographic requirements, rainfall parameters, preparedness content, and the functional boundaries of the proposed application and its Web Administration System.

In Bacoor City, flooding is a persistent and consequential reality. Survey results reveal that 92% of respondents have personally experienced flooding, with 58% describing the flooding as severe or waist-deep and higher. Despite this exposure, only 20% consider themselves well-informed about flood-prone areas. Mean satisfaction scores for information quality, timeliness, and clarity ranged from 2.62 to 2.72 on a 5-point scale, clustering at the lower end of the Neutral range and tending toward dissatisfaction. These information deficiencies have direct consequences: 64% of respondents reported making wrong decisions due to insufficient flood information, including taking flooded routes, failing to evacuate on time, and misjudging conditions. Open-ended responses further identified late or delayed information at 26% and insufficient or absent location-specific warnings at 20% as the leading concerns. How can an Android application using an Expert System provide residents with

understandable, location-specific flood susceptibility assessments based on expert-validated knowledge, rainfall scenarios, and geographic information?

The problem also concerns the accessibility and operational maintenance of institutional knowledge. The BDRRMO maintains historical flood records, exposure information, preparedness protocols, and evacuation-center information, while the CPDCO maintains geographic and planning data. Current dissemination practices, including social media, email, and barangay coordination, do not fully address the public's need for on-demand, situation-specific guidance. A community-facing platform must therefore communicate approved information while ensuring that rules, map objects, preparedness content, rainfall parameters, data sources, and administrative accounts can be maintained by authorized personnel. How can a Rule-Based Expert System, dynamic GIS-style map, Decision Support System, and Web Administration System organize this institutional knowledge into an explainable assessment and appropriate pre-event preparedness guidance for Bacoar City residents?

Taken together, these gaps reveal a disconnect between institutional flood knowledge and the decisions residents must make before flooding occurs. The overarching question of the study is: How can FloodSense integrate expert-provided and expert-validated knowledge, rule-based inference, scenario-based rainfall input, interactive geographic visualization, Check My Area, evacuation-center information, and structured decision support to strengthen flood susceptibility awareness and pre-event preparedness in Bacoar City, while remaining a complementary public-information tool that does not perform real-time flood prediction or supersede official warnings?

Objectives of the Study

Generally, the study aimed to develop an Android application that uses a Rule-Based Expert System to assess flood susceptibility and a Decision Support System to provide pre-event preparedness support for residents of Bacoar City through

explainable expert knowledge, rainfall scenarios, and dynamic geographic visualization.

Specifically, it aimed to:

1. identify the prevailing challenges, pain points, and needs of Bacoor City residents and relevant local government offices concerning flood information access, timeliness, clarity, location-specific awareness, and pre-event decision-making through a structured survey of 50 respondents across 12 barangays, a key informant interview with the Bacoor Disaster Risk Reduction and Management Office (BDRRMO) to validate survey findings and assess institutional knowledge and data availability, and a supplementary key informant interview with the City Planning and Development Coordinator Office (CPDCO) to identify available geographic, elevation, flood-hazard, land-use, and barangay-boundary information, all conducted during the Empathize phase of the Design Thinking methodology;
2. analyze expert-provided and expert-validated flood knowledge, BDRRMO historical flood records and preparedness protocols, CPDCO geographic information, PAGASA rainfall classifications, evacuation-center information, and user situational requirements to determine the facts, conditional rules, susceptibility levels, rainfall parameters, geographic entities, and DSS content required by the knowledge base and inference process;
3. develop an Android application using Flutter and Dart, supported by Django, Django REST Framework, GeoDjango, and PostgreSQL/PostGIS, that integrates a deterministic Rule-Based Expert System, scenario-based rainfall input, an interactive GIS-style map, manual location selection, optional one-time foreground location, explainable susceptibility output, and separate Decision Support

System guidance; provide a custom Web Administration System for authorized review and maintenance of permitted content and records; protect raw Expert System rules and algorithm logic from ordinary administrator editing; and establish a fixed-schema CSV exchange requirement for permitted values without representing that requirement as operational until its routes, validation, interface, integration, and tests are completed;

4. test the system's core modules by implementing unit testing, integration testing, system testing, rule and inference validation, application programming interface communication testing, spatial-query testing, database-operation testing, map interaction testing, DSS decision-flow testing, role-based administrative testing, and end-to-end validation to ensure that susceptibility assessment, geographic selection, dynamic visualization, preparedness guidance, and administrative data management operate accurately and consistently;
5. evaluate the system's overall level of user satisfaction in terms of accessibility, reliability, usability, security, maintainability, and portability of the system using ISO/IEC 25010.

Theoretical Framework

The theoretical framework of FloodSense system illustrates how a Rule-Based Expert System, a dynamic Geographic Information System map, and a separate Decision Support System are integrated into an Android and web platform to provide location-specific flood susceptibility information and pre-event preparedness support to residents of Bacoor City. Approved institutional data and expert-validated knowledge are managed through the Django Web Administration System and stored in PostgreSQL with PostGIS. The Flutter Android application communicates with the Django backend through secured application programming interfaces to request assessments, maps, explanations, preparedness guidance, and verified evacuation

resources. The framework keeps susceptibility classification, preparedness guidance, spatial visualization, and administrative publishing as coordinated but clearly defined functions.

Authentication and Onboarding Module. This module manages resident registration and login using an email address and password. The Django backend verifies credentials, applies password rules, supports email-based password recovery, and issues the session or token required for protected mobile requests. A user's local application state cannot grant administrator privileges because every protected action is checked on the server. During first launch, the onboarding screen explains the purpose of FloodSense, the rainfall controls, map legend, susceptibility meanings, Check My Area procedure, Decision Support System role, data date or version, privacy information, and safety disclaimer. The disclaimer emphasizes that the application provides scenario-based susceptibility and preparedness information rather than a live flood forecast or official warning.

PAGASA-Aligned Rainfall and Location Input Module. This module allows the resident to define the assessment scenario through plain-language rainfall intensity and duration selections. The baseline intensity labels are Light, Moderate, Heavy, Intense, and Torrential, while the supported durations may include 1, 3, 6, 12, and 24 hours. The internal thresholds and wording must follow the final PAGASA reference approved for the study. The resident may select a supported area from the map or manually place a pin through Check My Area. The selected rainfall values and location are planning inputs for the current request; they do not prove current weather conditions or the resident's actual physical location. Unsupported combinations produce a limitation response instead of an assumed classification.

Web Administration and Knowledge Management Module. This module is the controlled maintenance workspace used by authorized administrators. It manages geographic zones and map features, rainfall parameters, Expert System facts and rules, Decision Support System questions and branches, evacuation-center records,

public explanations and disclaimers, source and provenance information, and permitted user-support actions. Institutional materials from BDRRMO, CPDCO, PAGASA, and other approved providers are entered or imported as controlled records and are associated with their source, review status, effective date, and version. Critical changes follow the sequence Draft, Validation, Review, Approval, and Publication. Published versions remain available for audit or rollback, and incomplete or unapproved records are not released to residents.

Django REST and GeoDjango Processing Module. This module provides the controlled connection between the Android application, administrative workspace, processing services, and database. Django handles server-side authentication, permissions, validation, business logic, and administrative functions. Django REST Framework receives and returns secured JSON and GeoJSON data for the mobile application. GeoDjango provides the spatial models and operations used to store and query polygons and points. When Check My Area submits a coordinate, GeoDjango and PostGIS perform a point-in-polygon query and return the most specific supported zone. Neither the Android application nor the administrator's browser connects directly to the database; all operations pass through the controlled Django services.

Rule-Based Expert System Assessment Module. This module implements a deterministic Rule-Based Expert System. In rule-based expert-system theory, a knowledge base stores domain knowledge and an inference procedure applies rules to known facts to reach a conclusion (Buchanan & Shortliffe, 1984). FloodSense represents approved knowledge through geographic-zone facts, rainfall categories and durations, decision tables, structured IF-THEN rules, exceptions, priorities, sources, versions, and explanation text. Forward chaining begins with the resolved zone, rainfall scenario, active zone profile, and published rule set; it tests conditions and fires the applicable rules. The primary outputs are Low, Moderate, High, and Very High. Conflicting rules may return Uncertain, missing coverage may return Insufficient Data, and a point beyond approved boundaries returns Outside Supported Area.

Dynamic GIS Map and Check My Area Module. This module presents approved Bacoor City spatial data through a movable map rendered in Flutter using flutter_map. Residents can pan, zoom, tap supported zones, open information pop-ups, display map layers, read a text-supported color legend, view evacuation-center markers, and reposition the Check My Area pin. Supported polygons are colored Low in green, Moderate in yellow, High in orange, or Very High in red based on the current Expert System result. Changing the rainfall scenario causes the same approved polygons to be evaluated and recolored. Map precision is limited to the resolution of the approved boundary and zone data, and administrators cannot assign a public susceptibility class merely by painting a polygon color.

Results Summary and Explainable Result Module. This module presents the assessment in both summary and detailed form. The Results Summary lists supported areas with their current susceptibility badges and may allow searching or filtering by barangay or class. The detailed result identifies the resolved area, parent barangay, rainfall intensity and duration, susceptibility class, plain-language meaning, matched rule or decision-table basis, summarized conditions, source and version information, update date, and any limitation state. The same zone, scenario, and published rule version must produce the same result on the map, in Check My Area, and in the context given to the Decision Support System. A result is marked outdated when its scenario, location, or controlling version changes.

Decision Support System Module. This module provides pre-event preparedness guidance through a deterministic, database-driven decision tree. Decision support systems organize relevant information and structured guidance to assist human decision-making (Power, 2002). FloodSense begins with the latest valid Expert System result as read-only context and combines it with the resident's answers to approved situational questions. The configured path may provide guidance concerning go-bag preparation, family communication, protection of documents and valuables, assistance for vulnerable household members, monitoring of official

advisories, and evacuation readiness. The DSS is separate from the Expert System: it does not change the susceptibility class, predict live conditions, or issue an independent evacuation order.

Verified Evacuation Center Resource Module. This module displays only active evacuation-center records that have been verified from BDRRMO or another authorized city source. Each center is stored as a geographic point and may include its name, barangay, address, responsible office, verification status, validation date, and other information approved for public display. PostGIS can rank centers by straight-line geographic distance from the resident's selected pin, and the Android application can open the selected coordinates in an installed navigation application. A nearby ranking or external route does not guarantee that a road is safe or passable, that a center is currently open or has available capacity, or that the suggested route is appropriate during an actual emergency.

User Account and Preferences Module. This module allows authenticated residents to view or update approved profile and convenience information, including their display name, email address, home barangay, optional default rainfall scenario, password actions, privacy information, and logout. Passwords are securely processed by Django and are never shown to administrators in plain text. A manually placed Check My Area coordinate and DSS answers are used to complete the requested operation and are not retained as continuous location or behavior history by default. Resident accounts do not permit editing of Expert System rules, Decision Support System content, official map geometry, rainfall references, or evacuation-center records.

PostgreSQL and PostGIS Database Module. All authoritative relational and spatial records are stored in PostgreSQL with the PostGIS extension. The database contains users and roles, rainfall references, barangays and susceptibility zones, geographic profiles, evacuation-center points, Expert System rules and conditions, Decision Support System nodes and branches, source and review metadata,

published versions, and audit history. PostgreSQL enforces relational consistency, while PostGIS supports point-in-polygon and straight-line distance operations. The backend retrieves only the compatible active versions required for a request and returns only public approved data to residents. Internet access is required for fresh synchronization and server-side evaluation; any approved cached information must display its date and version when connectivity is unavailable.

Overall, the framework demonstrates how expert-validated knowledge, structured rule-based inference, geographic processing, explainable results, and a separate preparedness decision tree are combined in one community-accessible system. By converting approved institutional records and expert knowledge into a controlled mobile and web service, FloodSense aims to strengthen location-specific flood susceptibility awareness and support better pre-event preparation without replacing the authority of BDRRMO, PAGASA, barangay officials, or emergency responders.

The diagram presents FloodSense through a top-to-bottom organization. The Internet connection appears above the FloodSense system, the resident and administrator actors appear on the left, approved institutional sources and external display services appear on the right, and the authoritative PostgreSQL/PostGIS database appears below. The stacked modules inside the system boundary show the functions that process user inputs, institutional knowledge, susceptibility rules, spatial data, results, preparedness guidance, evacuation resources, and account information.

System Architecture

The system architecture uses a vertical client-server presentation of the approved FloodSense design. Resident requests enter through authentication, rainfall and location input, and the secured Django service layer. Institutional data and expert review enter through the Web Administration and Knowledge Management Module. The Rule-Based Expert System produces the susceptibility result, which is used by the map and explanation modules and passed as read-only context to the separate DSS.

The database stores the controlled relational and spatial records, while basemap and navigation services are used only for display and external directions.

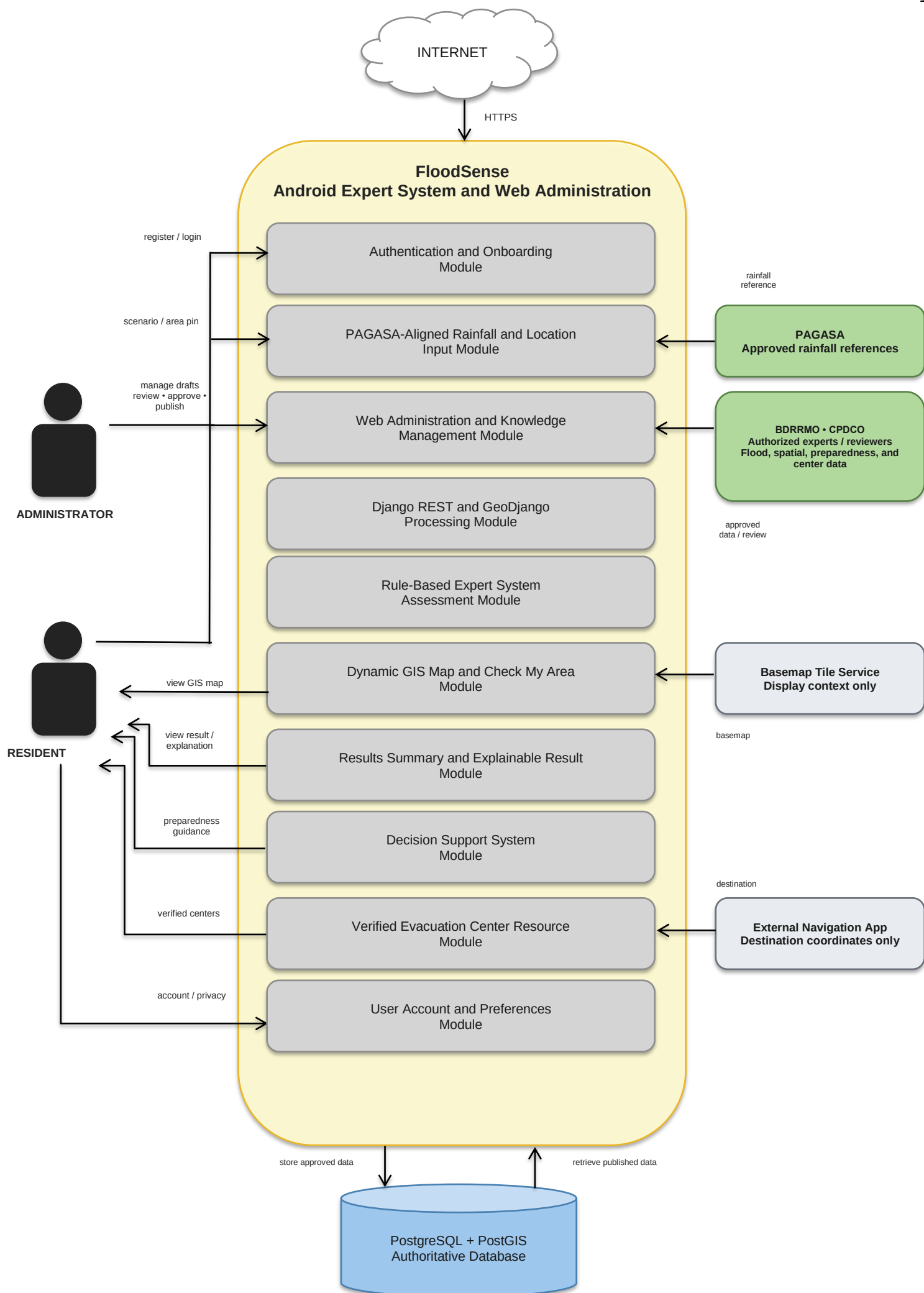


Figure 1. Revised System Architecture of FloodSense.

Significance of the Study

The development of FloodSense addresses a critical gap in flood information dissemination by translating expert-provided and expert-validated knowledge, institutional geographic information, and rainfall scenarios into an accessible susceptibility assessment, dynamic map, and pre-event preparedness support. The following stakeholders stand to benefit directly from the study.

For the Bacoor Disaster Risk Reduction and Management Office (BDRRMO). The study provides a prototype channel through which validated susceptibility information and pre-event preparedness guidance may eventually be communicated to residents. The custom Web Administration System currently supports controlled management of DSS guidance, evacuation-center records, source information, and audit history, together with read-only review of map, rainfall-reference, and parameter records. Ordinary administrators are not permitted to rewrite raw Expert System rules or inference logic. The proposed fixed-schema CSV workflow will permit only validated values within system-required fields after implementation and authorization. These functions complement rather than replace official advisories, warnings, and emergency directives.

For the City Planning and Development Coordinator Office - Bacoor (CPDCO). The study demonstrates a practical community use for geographic and planning information maintained by the office, including elevation, flood-hazard, land-use, and barangay-boundary data. The integration of verified geographic records into spatial polygons, point locations, and area-selection functions illustrates how institutional planning information can support public understanding of flood susceptibility without presenting FloodSense as a substitute for official technical maps.

For residents of historically flood-affected barangays. The Android application directly responds to the information deficits identified in the survey: 92% of respondents experienced flooding, only 20% felt well-informed about flood-prone areas, and 64% reported wrong decisions associated with insufficient information.

Residents can select a rainfall scenario, pan and zoom the dynamic map, inspect color-coded susceptibility polygons, use Check My Area, view verified evacuation-center locations, and receive preparedness guidance suited to the susceptibility result and their stated situation. Plain-language rainfall categories reduce the need to interpret technical meteorological measurements, while the disclaimer directs users to continue following BDRRMO and PAGASA advisories.

For future researchers. The study provides a locally grounded reference for combining knowledge engineering, rule-based inference, decision support, and geographic information in an Android disaster-preparedness application. The documented approach to expert knowledge acquisition, rule validation, spatial data management, dynamic map interaction, administrative maintenance, software testing, and ISO/IEC 25010 evaluation can inform related studies in other flood-prone communities, subject to validation with local experts, datasets, and user populations.

Time and Place of the Study

The research and development of the FloodSense application were conducted using the Design Thinking methodology, beginning with data-gathering activities before the formal study period. The official duration of the study spans from September 2026 to May 2027. Major activities, including requirements analysis, knowledge acquisition, rule structuring, system design, mobile and web development, geographic data preparation, testing, evaluation, and documentation, were carried out at the Department of Computer Studies, Cavite State University - Bacoar City Campus, located in Soldiers Hills IV, Molino, Bacoar City, Cavite. The campus served as the primary venue for software development, academic consultation, integration testing, and preparation of the research manuscript.

Initial data gathering commenced in early May 2026 during the Empathize phase of Design Thinking. A structured survey was administered to 50 residents across 12 barangays in Bacoar City, many of which have historically experienced flooding,

from May 5 to May 12, 2026, to capture prevailing challenges in flood information access, timeliness, clarity, and decision-making. On May 12, 2026, a key informant interview was conducted at the Bacoor Disaster Risk Reduction and Management Office (BDRRMO), located at the City of Bacoor Legislative and Disaster Resilience Building, Bacoor Government Center, Barangay Bayanan, Bacoor City. The interview aimed to validate survey findings, understand the office's workflows and information holdings, and assess the feasibility of a community-facing flood susceptibility and preparedness tool. A supplementary key informant interview was conducted on May 18, 2026 with the Officer-in-Charge of the Bacoor City Planning and Development Coordinator Office (CPDCO), located at the Bacoor Government Center, Barangay Bayanan, Bacoor City. The interview confirmed the availability of elevation maps, flood-hazard maps, land-use maps, and barangay-boundary maps and established the geographic context needed for the study. The outputs of the Empathize phase guided the identification of system requirements and the sources to be subjected to expert validation.

Following data gathering, the study proceeds through the Define, Ideate, Prototype, and Test phases within the research timeline. Requirements, interface wireframes, geographic data structures, susceptibility rules, DSS content, mobile screens, and Web Administration System functions are refined through consultations and development evidence. Final resident evaluation and system-quality analysis remain future activities. Results will be reported only after the approved instrument, sampling procedure, completed responses, and analysis are available.

Scope and Limitation of the Study

The study develops FloodSense as a scenario-based, pre-event flood-susceptibility assessment and preparedness-support system for Bacoor City. Its deterministic Rule-Based Expert System evaluates a user-confirmed rainfall scenario and supported location, while the separate Decision Support System presents

published preparedness guidance without changing the susceptibility result. The current prototype produces Low, Moderate, High, or Very High demonstration outputs and explanations. Operational use remains blocked until approved local susceptibility data, validated parameters, and institutional authorization are available. FloodSense does not provide real-time flood monitoring, official forecasts, warnings, evacuation orders, or safety guarantees.

The primary intended users of the Android application are Bacoor City residents. In the current mobile flow, a resident selects Light, Moderate, Heavy, Intense, or Torrential rainfall and a duration of 1, 3, 6, 12, or 24 hours, confirms the hypothetical scenario, selects or confirms a location, and explicitly requests an assessment. Resident registration, login, persistent profiles, and assessment history are not exposed in the current interface and remain unimplemented for the intended user flow.

The interactive map is limited to supported Bacoor City reference and demonstration data. It can pan and zoom, display scenario-dependent demonstration colors, accept manual area selection, and show a neutral 47-barangay administrative reference boundary. The boundary layer represents current barangay identities through a documented derivation and has passed technical geometry checks, but it is not automatically City-issued or City-verified and contains no flood-susceptibility facts. The map does not display live flood extent, water depth, traffic conditions, road closures, river level, center occupancy, or satellite rainfall imagery.

Location functions allow the user to select an area manually, place or adjust a map pin, or request the device's location once while the application is in the foreground. The application can propose a barangay from the coordinate and requires the user to confirm or correct it before assessment. The coordinate is used for the active request and is not continuously collected in the background. The current function does not find safe routes, verify road passability, or continuously track the resident.

The backend and custom Admin portal include an evacuation-center record workflow. This capability does not yet expose resident-facing nearby-center discovery, distance ranking, live capacity, operating status, or safe-route guidance. Any future resident display must rely on authorized and current records and must direct users to confirm emergency instructions through official channels.

The Decision Support System is confined to pre-event preparedness and remains separate from the susceptibility assessment. The current implementation can return published guidance associated with the assessment context. A resident-facing structured multi-question decision tree has not been verified as complete. The DSS does not alter the susceptibility class, issue autonomous evacuation orders, diagnose conditions, guarantee safety, or make the resident's final decision.

A separate custom Web Administration System provides authenticated access to a dashboard, read-only map-data and rainfall-reference review, parameter-governance views, and workflows for DSS content, evacuation-center records, data sources, and audit history. Ordinary administrators cannot directly edit raw Expert System rules, rule conditions, priorities, conflict-resolution logic, or the algorithm. Technical Django Admin remains a separate developer or research-maintenance interface. The confirmed fixed-schema CSV export-and-import workflow is not yet operational and will require routes, services, schema and value validation, interface controls, permissions, integration, and tests before it can be described as implemented.

The implementation stack uses Flutter and Dart for the mobile application; flutter_map for resident-facing mapping; Python Django and Django REST Framework for backend services and REST interfaces; GeoDjango for geographic operations; PostgreSQL with PostGIS for relational and spatial records; and Django templates with Leaflet for the custom administrative interface. The current prototype requires network access for integrated server functions. iOS support, fully offline operation, live sensors,

automated meteorological feeds, background polling, background GPS tracking, emergency dispatch, and machine-learning prediction are outside the current scope.

The reliability of any susceptibility assessment depends on the completeness, geographic resolution, provenance, institutional approval, and expert validation of the data and knowledge used. No approved operational susceptibility dataset is currently present in the repository. The available 47-barangay layer is an administrative reference only, while provisional research materials remain unvalidated or restricted. Demonstration results must therefore not be presented as official Bacoor City flood-hazard information.

Data privacy and security are addressed in accordance with Republic Act No. 10173, or the Data Privacy Act of 2012. The current location flow requests foreground permission only when the user chooses the location function, uses the coordinate for the active request, permits confirmation or correction of the detected barangay, and does not implement continuous background location tracking. Administrative functions require authenticated and authorized access. Evaluation findings will not be generalized until valid survey and system-evaluation data are available.

Definition of Terms

The following terms are defined operationally as they are used in this study to ensure clarity and consistency throughout the research:

Android Application. The resident-facing FloodSense software developed with Flutter. Its current interface supports scenario selection and confirmation, manual or one-time foreground location input, barangay confirmation or correction, assessment requests, map presentation, susceptibility results, explanations, and preparedness guidance. Resident accounts, assessment history, notifications, and offline operation are not currently exposed.

Check My Area. The location-selection flow that allows manual area selection, placement or adjustment of a map pin, and an optional one-time foreground location

request. A detected barangay is proposed for user confirmation or correction before assessment; the function does not continuously track the device.

Color-Coded Susceptibility Levels. The visual scheme used on the dynamic map in which green indicates Low, yellow indicates Moderate, orange indicates High, and red indicates Very High flood susceptibility for the active scenario.

CPDCO (City Planning and Development Coordinator Office). The Bacoor City office that maintains planning and geographic information, including elevation, land-use, flood-hazard, and barangay-boundary materials relevant to the study.

Decision Support System (DSS). The component separate from the Rule-Based Expert System that presents published pre-event preparedness guidance using the assessment context. It does not change the susceptibility result, issue an official warning or evacuation order, or make the user's final decision.

Dynamic Map. The interactive map that supports panning, zooming, scenario-dependent demonstration coloring, manual area selection, map-pin placement, and display of the neutral 47-barangay reference layer. The reference boundary is not an official flood-hazard dataset.

Expert-Provided and Expert-Validated Knowledge. Flood-related facts, parameters, rules, preparedness guidance, and supporting information that have been supplied or reviewed by an authorized source and recorded with adequate provenance. Repository presence or technical validation alone does not establish institutional approval.

Expert System. A knowledge-based computer system that applies stored facts and conditional rules through an inference process. In FloodSense, it produces an explainable susceptibility class for a supported geographic area and rainfall scenario.

Flood Susceptibility. The relative potential of a supported geographic area to experience rainfall-related flooding under a user-confirmed scenario. The current prototype demonstrates Low, Moderate, High, or Very High outputs but does not yet

contain an approved operational Bacoar City susceptibility dataset and is not a real-time prediction.

Geographic Information System (GIS). A system for organizing, analyzing, and presenting information associated with geographic locations. FloodSense uses GIS functions for polygons, points, containment queries, interactive visualization, and evacuation-center locations.

GeoDjango. The geographic extension of Django used by FloodSense to represent spatial data and perform geographic operations between backend services and the PostGIS database.

Inference Process. The procedure that matches facts from the selected rainfall scenario and geographic area against validated rules in the knowledge base to derive a susceptibility conclusion and supporting explanation.

Knowledge Base. The structured collection of expert-validated facts, conditional rules, geographic attributes, rainfall parameters, susceptibility categories, and related provenance used by the Expert System.

PAGASA-Aligned Rainfall Input. The current interface labels for Light, Moderate, Heavy, Intense, and Torrential rainfall and durations of 1, 3, 6, 12, or 24 hours. These selections are hypothetical scenario inputs. Their operational thresholds and use require authoritative source confirmation and validation before official deployment.

PostGIS. The spatial extension of PostgreSQL used to store and query FloodSense geographic polygons, points, boundaries, selected locations, and evacuation-center coordinates.

Preparedness Provenance. Information identifying the source, validation status, responsible actor, relevant date, and version of a preparedness item or related record. Provenance permits users and administrators to distinguish approved, provisional, and unsupported information.

Rainfall Parameters. The supported combination of rainfall intensity and duration selected by the user and applied as scenario facts during Expert System assessment.

Rule-Based Expert System. An Expert System in which domain knowledge is represented as conditional rules. FloodSense evaluates applicable rules against known scenario and geographic facts to produce a susceptibility assessment.

Web Administration System. The custom browser-based interface used by authorized administrators for the dashboard, read-only map and rainfall-reference review, parameter governance, DSS-content management, evacuation-center management, data-source management, and audit history. It does not expose raw rule editing. The separate technical Django Admin is reserved for developer or research maintenance. Fixed-schema CSV export and import is a confirmed but currently unimplemented requirement.

REVIEW OF RELATED LITERATURE

This chapter reviews literature and studies that clarify the concepts, technologies, responsibilities, and limitations relevant to FloodSense. The discussion is organized into foreign and local literature followed by foreign and local studies. It examines flood susceptibility, scenario-based assessment, deterministic Rule-Based Expert Systems, explainable inference, Decision Support Systems, geographic information, spatial databases, location privacy, risk communication, data provenance, administrative governance, software quality, and Philippine flood conditions.

The reviewed materials are used to explain the research problem and the design boundaries of FloodSense. Features described in another application or study are not treated as FloodSense functions. FloodSense remains a scenario-based, pre-event susceptibility-assessment and preparedness-support system in which the user confirms a hypothetical rainfall scenario and explicitly requests an assessment. It is distinct from operational forecasting, real-time monitoring, official warning, and automatic evacuation systems.

Related Foreign Literature

Flood Susceptibility and Scenario-Based Assessment

Flood susceptibility expresses the relative tendency of a location to experience flooding when relevant conditions are present. It differs from flood hazard, which concerns the probability and physical characteristics of an event; exposure, which concerns people or assets in affected places; vulnerability, which concerns the capacity to be harmed; and risk, which combines hazard and potential consequences. Membele, Naidu, and Mutanga (2022) found that flood-vulnerability mapping in developing countries varies substantially in terminology, indicators, spatial scale, data availability, and validation practice. Their review shows why a public system must name the type of assessment it performs and avoid presenting a general vulnerability or hazard map as a locally validated susceptibility conclusion.

Scenario-based assessment supports planning by holding selected conditions explicit instead of implying that the system is observing an event as it occurs. A hypothetical rainfall intensity and duration can be treated as assessment inputs only when their meanings, permitted combinations, and geographic applicability are documented. This supports the FloodSense interaction in which the resident selects or confirms a scenario before requesting assessment. It does not establish that the scenario is current, predicted, or officially issued, and it does not justify importing weights or thresholds derived for another place.

Deterministic Rule-Based Expert Systems and Knowledge Bases

A Rule-Based Expert System represents domain knowledge through facts, conditions, and production rules and applies an inference procedure to derive a conclusion. Buchanan and Shortliffe (1984) provide the foundational distinction between the knowledge base and the inference mechanism. More recently, Papadopoulos et al. (2022) reviewed technologies and standards used in rule-based decision-support systems and described how encoded knowledge and rules engines evaluate case-specific facts. Although their review concerns clinical systems, the architectural distinction is relevant to FloodSense: flood knowledge, scenario facts, geographic facts, and explanations belong to controlled knowledge structures, while the inference procedure determines which eligible rule applies.

Deterministic processing means that the same approved facts and the same published rule version should produce the same result. It does not imply that the rules are scientifically valid merely because the software executes them consistently. Knowledge acquisition, conflict handling, parameter definitions, version control, and expert or institutional validation remain necessary. Accordingly, ordinary FloodSense administrators may manage approved records and permitted values but do not rewrite raw rule conditions, priorities, conflict-resolution logic, or algorithm code. The

confirmed fixed-schema CSV exchange is a future controlled-value workflow and is not evidence that such exchange is currently operational.

Explainable Inference

Explainability is necessary when a system produces a classification that may influence preparedness decisions. Kostopoulos, Davrazos, and Kotsiantis (2024) reviewed explainable decision-support approaches and emphasized transparency, interpretability, and human-understandable justification. Their review primarily addresses artificial-intelligence-based systems, but its communication principles apply to deterministic inference: a user should be able to understand the relevant inputs, the basis of the result, and the limitations of the evidence without being required to inspect source code or raw rule definitions.

For FloodSense, an explanation should identify the confirmed rainfall scenario, resolved geographic context, susceptibility class, applicable basis, source or version information, and any limitation state. This supports accountability while preserving the distinction between explanation and rule administration. It also prevents a color or label from appearing authoritative without disclosing what it represents. Explainability does not remove the need for validated data, and a clear explanation cannot turn provisional information into an official local assessment.

Decision Support and Preparedness Guidance

A Decision Support System organizes information, alternatives, or recommended actions to assist human judgment. In flood preparedness, support may include understandable mitigation choices, household preparation, communication planning, protection of documents and valuables, assistance for vulnerable household members, attention to official advisories, and evacuation readiness. Mostafiz et al. (2022) observed that many web-based flood tools provide technical information but do not consistently provide locally actionable information for individuals and communities.

Their review supports combining understandable risk communication with practical guidance while clearly stating the tool's authority and limitations.

FloodSense separates the susceptibility assessment from preparedness guidance. The deterministic Expert System produces the assessment, while the DSS presents source-backed pre-event guidance without changing the susceptibility class. Information about evacuation facilities may support readiness when its source, verification date, location, and operational limitations are known, but distance alone does not prove route safety, current opening, accessibility, or available capacity. The DSS therefore assists preparation and does not issue an official warning, evacuation order, or guarantee of safety.

GIS, Interactive Maps, and Spatial Databases

Geographic Information Systems organize, analyze, and present information associated with locations. Their value in flood research depends on the suitability of the layers, coordinate reference systems, spatial resolution, classification methods, and validation procedures. An administrative boundary can resolve the name of a barangay or provide map context, but it contains no flood-susceptibility fact unless appropriate hazard or susceptibility evidence has been associated with it. This distinction is important when users may interpret any colored polygon as an official hazard layer.

Spatial databases support the controlled storage and querying of geographic records. The PostGIS Development Group (2026) explains that PostGIS extends PostgreSQL with geometry and geography types, spatial indexes, spatial relationships, coordinate transformation, validity operations, and spatial processing. These capabilities are relevant to point-in-polygon queries, administrative-area lookup, map-feature storage, and provenance-linked geographic records. They provide technical infrastructure, not scientific validation: a valid geometry and a successful spatial query do not establish that a flood conclusion is institutionally approved.

Flood-Risk Communication

Flood-risk information must be understandable, actionable, and explicit about uncertainty. Mostafiz et al. (2022) identified gaps in web-based flood communication, particularly where tools provide information without sufficiently connecting it to decisions that residents can make. Clear terms, legends, labels, explanation text, and source notices help prevent non-specialists from treating model outputs as certain predictions. Communication should also distinguish long-term or scenario-based information from time-sensitive official warnings.

These principles support a guided FloodSense flow rather than an unexplained technical map. Susceptibility colors must be paired with text, the selected scenario, and limitations. Preparedness information must direct users toward official channels during actual events. The interface must not imply that a neutral boundary layer, a demonstration color, or a provisional record represents a current flood condition.

Data Provenance and Geographic Validation

Provenance records where information originated, how it was transformed, which version was used, and what review or approval applies. Tullis and Kar (2021) connected provenance to ethical replicability and reproducibility in GIScience, including critical applications in which geographic data affect public decisions. Fakhruddin et al. (2022) similarly emphasized risk-informed data practices and the importance of understanding how disaster and climate data are produced, shared, interpreted, and governed.

For FloodSense, provenance should accompany susceptibility parameters, geographic layers, preparedness items, and resource information. Technical geometry validation may confirm coordinate-system consistency, valid shapes, or feature counts, but institutional validation concerns authority, currency, meaning, and permitted public use. The two forms of validation are related but not interchangeable. This distinction

prevents a technically processed dataset from being represented as official Bacoor City flood information without documented approval.

Foreground Location and Privacy

Mobile location can reveal a person's position and may contribute to identification or tracking. Android Developers (2026) distinguishes foreground access, one-time access, and background access and recommends requesting only the permission needed for the active function. This supports a FloodSense flow in which location is obtained only after a user initiates location selection or assessment, while manual selection remains available when the resident declines permission or prefers another area.

Foreground acquisition does not imply continuous monitoring. FloodSense does not require background GPS tracking, automatic polling, or autonomous location-based alerts. Location handling should remain purpose-limited to the active assessment, use only the precision necessary for the supported geographic operation, and avoid retaining unnecessary movement information. The presentation must also allow the user to confirm or correct the detected barangay because device accuracy and boundary proximity can affect geographic resolution.

Software Testing and Quality Evaluation

Software testing establishes whether specified behavior is implemented under defined conditions, while quality evaluation examines broader product characteristics. ISO/IEC 25010:2023 defines a product-quality model with nine characteristics for specifying and evaluating information and communication technology products (International Organization for Standardization [ISO], 2023). The model can guide the selection of criteria for FloodSense, but it does not provide a ready-made acceptance result or replace an approved evaluation instrument.

A defensible evaluation must identify the tested build, participants or evaluators, tasks, criteria, scale, procedure, valid responses, and interpretation method. Repository tests and interface demonstrations can support implementation verification, but they are not resident-acceptance findings. Because the final FloodSense survey and evaluation are not complete, Chapter 2 uses ISO/IEC 25010 only as literature supporting future assessment and does not report quality scores.

Related Local Literature

Philippine Flood Conditions and Institutional Preparedness

Official Philippine statistics continue to record substantial effects from extreme events and disasters. The Philippine Statistics Authority (PSA, 2026) summarizes national information on disaster occurrence, affected populations, deaths, and economic losses. These national figures establish the continuing relevance of disaster preparedness but do not identify the susceptibility of a particular Bacoor barangay or validate the inputs of a local application.

The National Disaster Risk Reduction and Management Council's National Disaster Response Plan identifies institutional responsibilities and coordinated actions for hydro-meteorological hazards (NDRRMC, 2024). The plan reinforces that warnings, response decisions, and emergency coordination belong to authorized government mechanisms. FloodSense may support pre-event understanding and preparedness, but residents must continue to follow PAGASA, local disaster offices, barangay authorities, and emergency responders during an actual event.

Official Hazard Information and Warning Authority

PAGASA operates rainfall and flood-information processes that use observations, forecasts, water-level information, and coordinated warning protocols. Its 2023 Annual Climate Bulletin explains the role of rainfall warning systems and the use of official warning levels for public action (PAGASA, 2024). FloodSense rainfall

selections are hypothetical assessment inputs; they are not automatic updates from PAGASA, an interpretation of current rainfall, or an official warning product.

GeoRisk Philippines presents HazardHunterPH as a national platform for indicative hazard assessment while advising users that results depend on available government data and do not replace official hazard reports from mandated agencies (GeoRisk Philippines, n.d.). Its source notices and limitations demonstrate responsible hazard communication. The same principle applies to FloodSense: sources, dates, limitations, and validation status should remain visible, and an indicative or provisional record should not be represented as an absolute prediction.

Cavite, Bacoar City, and the Imus River Basin

The Japan International Cooperation Agency reported the inauguration of flood-mitigation facilities associated with the Imus River Basin and low-lying areas of Imus and Bacoar (JICA, 2021). This institutional record confirms the relevance of watershed and infrastructure conditions to the local context. It does not provide barangay susceptibility classifications, rainfall-rule thresholds, or validation for FloodSense, and it should not be treated as a substitute for hydrologic or hazard data.

Administrative geography in Bacoar has also changed. The PSA documented the 2023 merger and renaming of barangays and currently lists 47 barangays in the City of Bacoar (PSA, 2023, 2025). These records support the current identities used in a reference layer. They do not establish the accuracy of every derived polygon, City verification of a third-party boundary, or flood conditions within any barangay. Administrative reference and flood-susceptibility evidence must therefore remain separate.

Administrative Data Governance and Responsible Use

Government publications, local records, and research datasets differ in authority and permitted use. An official list of administrative units may verify names

and codes, while an authorized hazard agency may provide hazard information, and a local office may validate operational records. Combining these sources requires documentation of ownership, date, scale, processing, approval, and restrictions. Provenance is particularly important when a dataset has been transformed, because technical processing can change geometry or attributes without changing the institution responsible for the original information.

FloodSense consequently distinguishes approved, provisional, technically validated, and unsupported information. The 47-barangay layer may serve as a current administrative reference after documented derivation and geometry checks, but it is not a flood-susceptibility layer and is not automatically City-issued or City-verified. No operational susceptibility dataset should be described as established until the necessary source, method, expert review, institutional authority, and database integration are documented.

Philippine Data Privacy and Software-Quality Context

Republic Act No. 10173, the Data Privacy Act of 2012, requires personal-information processing to observe transparency, legitimate purpose, and proportionality. For a resident-facing location function, these principles support clear notice, user initiation, limited collection, secure handling, and avoidance of continuous background tracking. They also support role-controlled administrative access. The Act is legally authoritative despite falling outside the preferred publication period for recent literature.

Philippine software studies have also used ISO/IEC 25010 as an evaluation framework, but the criteria and edition must be identified accurately. An evaluation of another application cannot be transferred to FloodSense, and the presence of a questionnaire does not establish completed results. FloodSense quality conclusions will require its own approved instrument, respondents or evaluators, documented procedure, and completed analysis.

Related Foreign Studies

Modelling on Assessment of Flood Risk Susceptibility at the Jia Bharali River Basin in Eastern Himalayas by Integrating Multicollinearity Tests and Geospatial Techniques. Debnath et al. (2024) investigated flood-risk susceptibility in the Jia Bharali River Basin in the Eastern Himalayas. The study used multiple flood-conditioning factors, multicollinearity testing, geospatial preparation, and several modeling approaches to classify susceptibility. The researchers compared the models and reported validation results using available flood evidence. The geographic coverage, conditioning factors, and hydrological records were specific to the basin, and the authors identified data availability, resolution, and long-term records as important limitations.

The study is similar to FloodSense in treating susceptibility as a classification that depends on explicit geographic and environmental information and in recognizing the need for validation. It differs because FloodSense uses deterministic scenario rules intended for a guided resident interaction rather than the study's geospatial models. None of the study's factor values, model performance, or susceptibility classes can be transferred directly to Bacoor City. Its contribution is methodological: it demonstrates that a classification requires documented variables, preparation, comparison, and validation before public use.

Assessment of the Susceptibility of Urban Flooding Using GIS with an Analytical Hierarchy Process in Hanoi, Vietnam. Nguyen, Fukuda, and Nguyen (2024) assessed urban flood susceptibility in Hanoi using a Geographic Information System and the analytical hierarchy process. The study combined nine conditioning factors, expert judgment, spatial normalization, weighted overlay, and historical flood points. Its output was a five-level susceptibility map evaluated through receiver operating characteristic analysis. The method was designed for Hanoi's physical and

urban conditions and depends on the selected factors, expert comparisons, source data, and validation points.

The study supports FloodSense requirements for explicit parameters, traceable expert judgment, spatial processing, and validation. Both approaches communicate ordered susceptibility categories, but FloodSense does not currently apply AHP or reuse Hanoi's factor weights. Hanoi's geography, drainage, land cover, and evidence differ from Bacoor City, so its weights and performance measures cannot justify FloodSense classifications. The study instead illustrates why local experts and evidence must review any parameter set before it becomes operational.

Flood Risk Communication Using ArcGIS StoryMaps. Oubennaceur, Chokmani, El Alem, and Gauthier (2021) examined the use of ArcGIS StoryMaps to communicate flood-risk information to non-specialists. The system combined maps, text, visual explanation, and interactive exploration to make technical content more accessible. The study emphasized understandable legends, careful category presentation, limited technical language, and contextual explanation. Its purpose was communication rather than deterministic rule-based assessment, and it did not provide data or classifications for Bacoor City.

The study is similar to FloodSense in using an interactive map and explanatory content for public understanding. FloodSense differs by asking the user to confirm a hypothetical rainfall scenario and by producing a rule-based susceptibility result before presenting separate preparedness guidance. The StoryMaps design principles can support legibility and explanation, but its map content, evaluation setting, and platform functions cannot be treated as evidence that FloodSense data are validated or that its outputs are official warnings.

Flood Mitigation Data Analytics and Decision Support Framework: Iowa Middle Cedar Watershed Case Study. Alabbad, Yildirim, and Demir (2022) developed a web-based decision-support framework for the Iowa Middle Cedar

Watershed. The framework integrated geographic and property information, flood scenarios, mitigation alternatives, and decision-tree analysis to assist users in considering mitigation options. The study demonstrates how maps, scenarios, analytics, and structured recommendations can be combined while leaving the final decision to the user.

The framework supports the FloodSense concept of separating analytical results from guidance. FloodSense, however, focuses on a resident-confirmed pre-event scenario and deterministic susceptibility assessment for supported Bacoar locations, followed by preparedness guidance. It does not copy the Iowa property data, mitigation calculations, decision trees, or geographic assumptions. The study also does not authorize FloodSense to issue evacuation orders or forecasts; its relevance lies in organizing evidence and alternatives for human judgment.

A Service-Oriented Collaborative Approach to Disaster Decision Support by Integrating Geospatial Resources and Task Chain. Fang et al. (2023) investigated a service-oriented approach for integrating geospatial resources and task chains in disaster decision support. The approach separated data resources, processing services, workflow tasks, and user-facing decision functions so that complex operations could be coordinated and traced. The study addressed a broad disaster-management environment rather than a single resident-facing flood-susceptibility application.

The separation of resources, processing, and presentation is relevant to FloodSense's mobile client, REST services, spatial data, deterministic inference, and administrative workflows. The systems differ in scale, users, and operational purpose. FloodSense does not implement the study's collaborative task-chain environment, and the study supplies no Bacoar-specific hazard facts. Its contribution is the architectural principle that geographic data and decision services should remain connected through traceable interfaces rather than being mixed into an unexplained display.

An Impacts-Based Flood Decision Support System for a Tropical Pacific Island Catchment with Short Warnings Lead Time. Williams et al. (2021) described a pilot impacts-based flood decision-support system for the Vaisigano River catchment in Samoa. The system combined observed and forecast rainfall, water-level information, hydrological thresholds, impact modeling, and web-supported geographic outputs for agencies responsible for warnings and response. It was designed for short lead times and operational decision support, with institutional participation and catchment-specific modeling.

This study is useful primarily as a contrast. Both systems seek to present understandable flood-related information, but the Samoa system depends on monitoring, forecasts, thresholds, and warning operations that FloodSense does not possess. FloodSense must not claim real-time monitoring, automatic rainfall updates, official warnings, or autonomous alerts. The catchment thresholds, impact models, data feeds, and institutional procedures cannot be transferred to Bacoar. The comparison clarifies why FloodSense remains a user-requested scenario assessment rather than an early-warning system.

Related Local Studies

High-Resolution Urban Change Modeling and Flood Exposure Estimation at a National Scale Using Open Geospatial Data: A Case Study of the Philippines. Johnson et al. (2021) investigated urban change and potential flood exposure across the Philippines using open geospatial data and national-scale modeling. The study combined land-change analysis with flood-exposure estimation to examine how urban development may increase the number of people and built-up areas exposed to flooding. Its intended users include planners and researchers who require broad evidence for national and regional policy.

The study supports FloodSense's use of geographic processing and its insistence on documenting source, scale, and uncertainty. It differs because national

exposure modeling is not a barangay-level scenario assessment and does not provide a deterministic resident-facing rule set. Its resolution, prospective assumptions, and national coverage limit direct application to Bacoor. The study's values and flood layers cannot be imported as approved local susceptibility data without examining their scale, currency, licensing, method, and fit with Bacoor conditions.

Developing an Integrated Multiscale Rainfall-Runoff and Inundation Model: Application to an Extreme Rainfall Event in Marikina-Pasig River Basin, Philippines. Dasallas, An, and Lee (2022) developed an integrated multiscale rainfall-runoff and inundation model for an extreme rainfall event in the Marikina-Pasig River Basin. The study connected hydrologic and two-dimensional inundation modeling and used terrain and event information to reproduce flood behavior at different spatial scales. Its results demonstrate the data, calibration, computational processing, and validation needed for physically based event modeling.

The study is relevant to FloodSense because it shows that flood depth and extent cannot be inferred from rainfall labels alone. It differs fundamentally from FloodSense's deterministic scenario classification and pre-event guidance. The Marikina-Pasig basin, event conditions, terrain, calibration, and hydraulic parameters cannot be transferred to Bacoor City. The study therefore supports a limitation statement: FloodSense should not describe its assessment as an inundation forecast, water-depth estimate, or real-time hydraulic model.

Flash Flood Modeling in the Data-Poor Basin: A Case Study in Matina River Basin. Macalalad et al. (2021) examined flash-flood modeling in the Matina River Basin in Davao City, where observation and calibration data were limited. The researchers combined available rainfall, basin, and modeling information to simulate a documented flood event and examined performance under data-poor conditions. The study directly addresses a problem shared by many Philippine localities: the scientific

model may be constrained by incomplete monitoring records and uneven data coverage.

The study supports FloodSense's use of explicit insufficient-data and limitation states rather than an assumed classification. Both recognize that a technical method is only as defensible as its source data and validation. FloodSense differs because it is not a hydrologic simulation model and does not estimate discharge, water level, depth, or flood extent. The Matina calibration and event results cannot be generalized to Bacoar, but the study supports conservative communication when local evidence is incomplete.

GIS-Based Identification of Flood Risk Zone in a Rural Municipality Using Fuzzy Analytical Hierarchy Process. Gacul et al. (2024) developed a GIS-based flood-risk zoning study for the Municipality of Santa Fe, Romblon. The researchers used government and online geospatial datasets, expert judgments, fuzzy analytical hierarchy process weighting, and geographic overlay to identify areas with different levels of flood risk. The work was a research model and mapping study intended to support local planning; it was not reported as a continuously operating public warning application.

The study resembles FloodSense in organizing geographic factors into understandable categories and in recognizing local-government relevance. It differs in method, rural setting, factors, expert group, and output. FloodSense does not use FAHP weights and must not transfer Santa Fe's classifications to Bacoar. The authors' concerns about current data, high-resolution terrain, and expert judgment support FloodSense's requirements for source documentation, local review, and geographic validation before public release.

Vulnerability Profile and Risk Perception towards an Inclusive Disaster Risk Reduction for the Flood Vulnerable Communities of Davao City. Cayamanda, Paunlagui, Bacongus, and Quimbo (2021) studied vulnerability and risk

perception in five flood-vulnerable Davao City communities. The research involved 353 respondents and used survey data together with key-informant and group-based qualitative evidence to examine social conditions, awareness, experience, and disaster-risk-reduction needs. Its focus was community vulnerability and inclusive preparedness rather than the computation of physical flood susceptibility.

The study supports FloodSense's attention to understandable communication, household preparedness, and the varied needs of residents. It differs because FloodSense does not calculate the social-vulnerability profile developed in the Davao research, and the respondents do not represent Bacoar residents. The study's social findings cannot be converted into susceptibility rules. Its transferable concept is that technical information must be paired with communication and preparedness support that residents can understand and use.

A Systematic Framework for Assessing Social Vulnerability to Flood for Integrated Flood Risk Management: A Case Study in Metro Manila, Philippines.

Dulawan, Imamura, Konishi, Amaguchi, and Ohara (2024) developed and applied a social-vulnerability framework in Manila, Marikina, and Paranaque. The study combined household survey information, census and planning data, exposure, susceptibility, and resilience indicators, analytical hierarchy process weighting, simple additive weighting, validation against historical damage, robustness checks, and scenario analysis. It was intended to support planners and integrated flood-risk management.

The study demonstrates that vulnerability is multidimensional and context-dependent. FloodSense is similar in requiring explicit factors, traceability, validation, and careful communication, but it does not calculate the study's social-vulnerability index. The three Metro Manila cities differ from Bacoar in population, infrastructure, hazards, and available data. Their indicators, weights, and city rankings cannot be transferred directly. The study supports local surveys and validation when social

information is included, while FloodSense currently limits its assessment to its approved susceptibility method and separate preparedness guidance.

An Expanded Interpretive Structural Modeling Analysis of the Barriers to Integrated Flood Risk Management Adaptation in Metro Manila. Mercado, Kawamura, and Medina (2023) examined relationships among barriers to integrated flood-risk-management adaptation in Metro Manila. The researchers used an expanded interpretive structural modeling method to arrange direct and indirect relationships among institutional, technical, financial, information, and coordination barriers. The study is a governance and planning analysis rather than a resident application or flood-susceptibility model.

Its relevance to FloodSense lies in recognizing that flood information systems operate within institutional responsibilities, data limitations, coordination requirements, and public trust. FloodSense similarly requires documented sources and appropriate validation, but it does not implement the study's barrier model. Metro Manila governance relationships cannot be assumed for Bacoar without local evidence. The study supports treating institutional approval and administrative governance as substantive research constraints rather than merely software configuration.

Web-Based Solution for Flood Warning Decision Support in the Province of Leyte, Philippines. Bentoso, Juan, Brosas, Paragas, Nuevas, and Velarde (2021) presented a web-based flood-warning decision-support prototype for Leyte. The conference paper described strategically positioned rainfall and water-level sensors, data transmission through communication networks, graphical web presentation, alert messages, and management support for monitoring flood-prone areas. The reported work concerned a proposed and tested automated warning solution; the available evidence does not establish province-wide operational deployment.

The system is similar to FloodSense in using a digital interface to make flood-related information understandable and in supporting preparedness decisions. It differs

in its reliance on sensors, real-time readings, monitoring, warning levels, and alerts. FloodSense has none of those capabilities and must not inherit them through comparison. Leyte's sensor locations, warning thresholds, communications infrastructure, and evaluation evidence cannot be transferred to Bacoor. The comparison instead clarifies the boundary between operational monitoring and FloodSense's user-requested hypothetical scenario assessment.

Assessment of SIAS Application Using Software Quality Model. Gumba, Brosas, and Paragas (2021) evaluated a Philippine Student Information and Accounting System using an online questionnaire based on ISO/IEC 25010. The system served educational institutions and was assessed using the eight product-quality characteristics in the 2011 edition of the standard. The study reported that the evaluated application satisfied its chosen criteria, but its participants, tasks, software domain, and version of ISO/IEC 25010 were specific to that application.

The study is relevant because it demonstrates a Philippine use of a structured software-quality model. It does not validate FloodSense, and its scores, questionnaire, respondents, or conclusion cannot be transferred. FloodSense must use the current ISO/IEC 25010:2023 model and its own approved evaluation procedure. Until that evaluation is completed, the standard remains an evaluation basis rather than evidence of acceptance or production readiness.

Synthesis of the Reviewed Literature

The literature agrees that flood-related classifications depend on the meaning, quality, scale, and validation of their inputs. Susceptibility maps, hydraulic models, exposure estimates, and social-vulnerability indices answer different questions. They cannot be treated as interchangeable merely because all use maps or ordered categories. For FloodSense, this requires a precise description of scenario-based susceptibility and prevents administrative boundaries, national exposure products, social indices, or another basin's model from being represented as local flood truth.

The reviewed Rule-Based Expert System and explainability literature supports the separation of a controlled knowledge base from its inference procedure. Deterministic rules can produce traceable results when facts, conditions, priorities, versions, and explanations are documented. However, deterministic execution is not a substitute for expert and institutional validation. Protecting raw rule logic from ordinary administration and limiting future CSV exchange to permitted values preserve the difference between governed data maintenance and rewriting the assessment algorithm.

Decision-support and risk-communication research further shows that a classification alone is insufficient for public use. Users require understandable labels, limitations, sources, and actionable preparedness information. FloodSense therefore separates susceptibility assessment from DSS guidance. The guidance may assist household preparation and encourage attention to official channels, but it does not change the assessment, predict current conditions, issue a warning, or order evacuation.

GIS and spatial-database technologies support location selection, containment queries, storage, map visualization, and geographic processing. The Philippine studies demonstrate that similar technologies can support national exposure assessment, basin modeling, municipal risk zoning, community research, and warning prototypes. Their diversity also demonstrates why external formulas, weights, thresholds, and system functions cannot be transferred automatically. Each method reflects a different geographic scale, dataset, user group, and institutional purpose.

The local literature emphasizes the authority of PAGASA, NDRRMC, GeoRisk Philippines, PSA, local disaster offices, and other custodians. It also shows that Bacoor's current 47-barangay identity structure is an administrative matter, not a flood-susceptibility dataset. Technical geometry checks and provenance records are necessary, but public operational use still requires appropriate local and institutional validation.

Finally, the software-quality literature supports systematic testing and evaluation but does not permit results to be claimed before data collection and analysis are complete. Automated tests can demonstrate specified implementation behavior, while an approved ISO/IEC 25010-based evaluation can examine broader product quality. FloodSense must report each type of evidence according to what was actually tested and must not treat an empty survey package as completed evaluation.

Research Gap

Existing research already provides sophisticated flood-susceptibility maps, hydraulic and inundation models, national exposure estimates, social-vulnerability frameworks, interactive risk-communication products, decision-support frameworks, and operational monitoring or warning prototypes. These contributions establish methods and design principles that FloodSense can learn from. They also demonstrate that real-time warning and forecasting require observations, calibrated models, thresholds, communications infrastructure, and institutional authority that are outside the current FloodSense scope.

The reviewed literature provides limited evidence for a Bacoar-specific public interaction that combines a user-confirmed hypothetical rainfall scenario, a supported location, deterministic and explainable rule matching, and separate pre-event preparedness guidance while making provenance and authority limits visible. This is not a claim that such a combination has never existed. It identifies an insufficiently documented intersection within the sources reviewed, especially for Bacoar City and its current barangay structure.

A more significant gap concerns local validation. The reviewed Philippine studies were conducted at national scale or in Metro Manila, Davao, Romblon, Leyte, and other basins. Even the Cavite institutional material does not provide approved FloodSense susceptibility parameters. Their findings clarify methods, communication needs, and governance problems but cannot establish Bacoar classifications. Local

data, expert review, source permission, institutional validation, and transparent versioning remain necessary before FloodSense can be represented as an operational public assessment.

FloodSense addresses the documented design gap as a bounded research prototype: the resident confirms a hypothetical scenario and requests an assessment; the deterministic Expert System produces an explainable susceptibility result from approved inputs; and the separate DSS presents source-backed preparedness guidance. FloodSense remains different from PAGASA products, official hazard reports, real-time warning platforms, monitoring networks, and evacuation authorities. Its value depends on maintaining that distinction and reporting unsupported data or functions as limitations rather than capabilities.

Comparative Synthesis of Related Studies

The following tables summarize the principal relationships among the reviewed studies. The comparisons identify method, contribution, limitation, and relevance without treating any external dataset, formula, or feature as part of FloodSense.

Table 1. Comparison of related foreign studies

STUDY/SYSTEM	AUTHOR/YR	METHOD	MAIN VALUE	LIMIT	FLOODSENSE LINK
Jia Bharali susceptibility modeling	Debnath et al., 2024; India	GIS and model comparison	Explicit factors and validation	Basin-specific data	Supports validation; parameters not transferable
Urban flood susceptibility in Hanoi	Nguyen et al., 2024; Vietnam	GIS and AHP	Traceable expert weighting	Hanoi-specific weights	Supports factor documentation; FloodSense does not use AHP
Flood Risk Communication Using ArcGIS StoryMaps	Oubennaceur et al., 2021; Canada	Interactive risk communication	Understandable maps and explanations	Communication , not inference	Supports map clarity; no Bacoar data
Flood Mitigation DSS	Alabbad et al., 2022; United States	Web DSS and decision tree	Scenario-based alternatives	Different data and users	Supports separate guidance; not an official order
Service-oriented disaster DSS	Fang et al., 2023; international	Geospatial services and task chains	Traceable separation of resources	Broader operational scope	Supports architecture; no Bacoar hazard facts

STUDY/SYSTEM	AUTHOR/YR	METHOD	MAIN VALUE	LIMIT	FLOODSENSE LINK
Impacts-Based Flood DSS	Williams et al., 2021; Samoa	Monitoring, forecasts, and impacts	Operational early-warning support	Catchment-specific infrastructure	Clarifies functions outside FloodSense scope

Table 2. Comparison of related local studies and systems

STUDY/SYSTEM	AUTHOR/YR	METHOD	MAIN VALUE	LIMIT	FLOODSENSE LINK
National urban change and flood exposure	Johnson et al., 2021; Philippines	Open geospatial modeling	National exposure evidence	Not barangay susceptibility	Supports scale and provenance controls
Marikina-Pasig rainfall-runoff and inundation model	Dasallas et al., 2022; Metro Manila	Hydrologic and 2D model	Event-based inundation modeling	Different basin and event	Shows why FloodSense is not a depth forecast
Matina flash-flood modeling	Macalalad et al., 2021; Davao	Data-poor basin model	Documents data limitations	Limited local observations	Supports insufficient-data states
Santa Fe flood-risk zoning	Gacul et al., 2024; Romblon	GIS and FAHP	Local expert-weighted mapping	Rural and location-specific	Supports local review; weights not transferable
Davao vulnerability and risk perception	Cayamanda et al., 2021; Davao	Survey and qualitative study	Community preparedness context	Not physical susceptibility	Supports understandable guidance

Table 2. Comparison of related local studies and systems (Continued)

STUDY/SYSTEM	AUTHOR/YR	METHOD	MAIN VALUE	LIMIT	FLOODSENSE LINK
Metro Manila social-vulnerability framework	Dulawan et al., 2024; Metro Manila	AHP, SAW, survey, and validation	Multidimensional local framework	Three-city social index	Supports context-sensitive validation
Metro Manila IFRM barriers	Mercado et al., 2023; Metro Manila	Expanded ISM	Institutional governance relationships	Not an assessment system	Supports approval and coordination boundaries
Leyte web flood-warning solution	Bentoso et al., 2021; Leyte	Sensors, web monitoring, and alerts	Real-time warning prototype	Deployment not established	Clarifies monitoring and alerts outside scope
SIAS software-quality assessment	Gumba et al., 2021; Philippines	ISO/IEC 25010 survey	Local quality example	Different software and 2011 model	Supports evaluation method, not results

Table 3. Comparison of FloodSense with selected existing systems and approaches

SYSTEM	BASIS	OUTPUT	TIME/SCOPE	SIMILARITY	DIFFERENCE
HazardHunterPH	Available government hazard datasets	Indicative hazard report	National reference platform	Public geographic information	Not FloodSense's scenario rule assessment
Leyte flood-warning prototype	Rain and water-level sensors	Monitoring display and alerts	Real-time warning purpose	Digital flood information	FloodSense has no sensors or autonomous alerts
Samoa impacts-based DSS	Observed and forecast hydro-meteorological data	Impact and warning support	Agency operational context	Decision-support communication	FloodSense is not an early-warning system
GIS/AHP susceptibility studies	Weighted spatial factors	Susceptibility map	Research-specific datasets	Ordered geographic classes	FloodSense uses protected deterministic rules, not transferred AHP weights
FloodSense	User-confirmed hypothetical rainfall and supported location	Explainable susceptibility plus separate guidance	Pre-event research prototype	GIS and decision support	No official forecast, warning, monitoring, or automatic order

METHODOLOGY

This chapter presents the research and development method for FloodSense, the current system design supported by the repository, the completed requirements-gathering activities, and the procedures planned for testing and evaluation. Completed functions are described in the present or past tense. Ongoing, planned, confirmed but unimplemented, and data-dependent work is identified accordingly. No survey result, software-quality score, acceptance result, or statistical finding is reported without completed and validated evidence.

Research Design

The study uses a developmental research design to produce and examine a mobile and web-based information system. Design Thinking organizes the research activities into Empathize, Define, Ideate, Prototype, and Test. The approach links resident information needs and institutional evidence to an iterative prototype while allowing requirements, interfaces, and validation boundaries to be refined as evidence becomes available.

The initial Empathize activity included the survey of 50 residents from 12 Bacoor City barangays and key-informant interviews described in Chapter 1. These completed activities are used as requirements evidence. The newer survey package is treated only as a research instrument because it does not contain validated responses or analysis.

Research Setting and Participants

The research is conducted at Cavite State University - Bacoor City Campus, with Bacoor City as the geographic and institutional context of the application. The completed requirements evidence involves the 50 initial resident respondents and the available BDRMO and CPDCO key-informant records identified in Chapter 1. Their

contributions concern information needs, institutional context, and potential data sources; they do not by themselves validate every current feature or dataset.

Participants for the final resident survey and system evaluation will be reported only after the sampling method, inclusion criteria, valid response count, and completed responses have been approved and verified.

[AUTHOR NOTE: Survey data collection is ongoing. Complete this section after the validated survey responses and approved analysis are available.]

Research Instruments

The completed requirements-gathering instruments consist of the initial resident survey and the available key-informant interview guides and records. Repository documentation, approved school documents, current source code, database migrations, interfaces, and automated tests are used as development evidence. Plans, mockups, diagrams, screenshots, and provisional datasets are used only according to their stated status and are not treated as proof of an implemented or approved feature.

The newer survey questionnaire and evaluation materials remain instruments rather than results. Before use in final analysis, their validity, administration procedure, respondent eligibility, response completeness, and statistical treatment must be documented.

Development Procedure

Empathize

The researchers identify resident information problems and institutional practices through the completed initial survey, interviews, and document review. The activity focuses on access to location-specific susceptibility information, clarity and

timing of preparedness information, and the institutional sources needed for responsible geographic and preparedness content.

Define

The requirements are translated into a bounded problem definition: a resident selects or confirms a hypothetical rainfall scenario and a supported location, requests an assessment, receives a deterministic susceptibility result with an explanation, and views separate pre-event preparedness guidance. Real-time monitoring, official forecasting, autonomous alerts, background tracking, automatic evacuation orders, live center occupancy, and safe-road routing are excluded.

Ideate

Alternative mobile interactions, map behaviors, result presentations, administrative workflows, and data-governance controls are represented in requirements documents and interface designs. Each proposed element is checked against the research objective, the privacy boundary, data availability, and the separation between susceptibility assessment and DSS guidance.

Prototype

The prototype is implemented through a Flutter mobile application, a Django and Django REST Framework backend, PostgreSQL/PostGIS spatial storage, and a custom web Admin portal. The prototype is refined by comparing mobile screens, routes, services, models, migrations, permissions, templates, and tests with the documented requirements.

Test and Iterate

Testing and iteration cover input validation, deterministic inference, rule matching and conflict handling, geographic lookup, API behavior, permissions, mobile interaction, custom Admin workflows, and appropriate limitation responses. Final acceptance and quality conclusions will be added only after a complete, dated test run and the approved user evaluation are available.

[DIAGRAM PLACEHOLDER: Insert the approved Design Thinking or system-development process diagram here.]

System Design and Current Implementation

Mobile Application

The Flutter mobile application implements rainfall-intensity choices of Light, Moderate, Heavy, Intense, and Torrential and duration choices of 1, 3, 6, 12, and 24 hours. The user confirms the scenario before requesting an assessment. The location flow supports manual selection, a movable map pin, and an optional one-time foreground GPS request. A detected barangay is proposed for user confirmation or correction. The application does not implement continuous background tracking.

The mobile map supports panning, zooming, demonstration scenario coloring, manual selection, and display of the neutral 47-barangay reference layer. The current layer is an administrative identity reference produced through a documented derivation. Its geometry has been technically checked, but it is not established as City-issued or City-verified and contains no flood-susceptibility facts.

Backend, REST Services, and Spatial Database

Django and Django REST Framework provide validation, permissions, APIs, business logic, and administrative services. GeoDjango and PostgreSQL/PostGIS support spatial records and point-to-area processing. Models and migrations provide database structure, but a model is not treated as a complete user function unless the relevant route, service, interface, integration, permissions, tests, and required data are also present.

Deterministic Rule-Based Expert System

The Expert System uses deterministic rule matching for a confirmed rainfall scenario and supported geographic context. It produces a susceptibility class and explanation. The backend includes rule matching, priorities, conflict handling,

validation, and related tests. The system does not use machine-learning prediction, automatic retraining, or self-modifying rules.

Decision Support System

The DSS is maintained separately from assessment logic. Its current backend and Admin workflow manage preparedness content that can be published and returned as guidance. The DSS consumes the assessment context but does not change the susceptibility class. A complete resident-facing structured multi-question flow has not been verified and remains developing or planned.

Custom Admin Portal and Technical Django Admin

The custom Admin portal implements authentication, a dashboard, read-only map-data review, rainfall-reference review, parameter-governance views, DSS-content management, evacuation-center management, data-source management, and audit-history review. Ordinary administrators do not directly edit raw rules, rule conditions, priorities, conflict resolution, or the inference algorithm. Technical Django Admin is a separate maintenance surface for developer or research use and is not the ordinary administrator workflow.

Fixed-Schema CSV Requirement

The confirmed design requires the administrator to export a canonical CSV table with system-required column names, parameter identifiers, and data types. Authorized editors may change only permitted values within the fixed fields. Import must validate the structure, required fields, parameter identifiers, values, and data types and reject incompatible files. This workflow will not expose raw rule editing or permit algorithm changes.

CSV export and import is a confirmed requirement but is not implemented in the current repository. It will remain described as future implementation until routes, services, validation, interface controls, permissions, integration, and tests are present and verified.

Data Sources, Provenance, and Validation

The backend and custom Admin portal support source records, validation status, and audit history. The approved-data directory currently contains guidance but no approved operational susceptibility dataset. Provisional data remains fictional, incomplete, unvalidated, restricted, or awaiting approval. Processing scripts demonstrate that a preparation or verification method exists, but they do not prove institutional approval or operational database use.

[DIAGRAM PLACEHOLDER: Insert the approved context diagram here.]

[DIAGRAM PLACEHOLDER: Insert the approved data-flow diagram here.]

[DIAGRAM PLACEHOLDER: Insert the approved use-case diagram here.]

*[DIAGRAM PLACEHOLDER: Insert the approved entity-relationship diagram
here.]*

Current Development Status

Table 2. Summary of Current Prototype Status

FUNCTION	STATUS	METHODOLOGICAL TREATMENT
Scenario selection, confirmation, deterministic assessment, explanation	Implemented and currently verified in code	Included in prototype testing
Manual selection, map pin, foreground GPS, barangay confirmation	Implemented and currently verified in code	Verify permissions, correction, and limitation behavior
Custom Admin dashboard and governed content workflows	Implemented and currently verified in code	Test role permissions and workflow validation
Fixed-schema CSV export and import	Confirmed requirement - not yet implemented	Future implementation and testing
Resident nearest-center discovery	Not implemented	Do not evaluate as an available resident function
Resident accounts, assessment history, notifications, offline use	Not implemented or outside current scope	Exclude from current effectiveness claims
Approved operational susceptibility data	Blocked or awaiting data	Demonstration output only; no official claim
Final survey and ISO/IEC 25010 evaluation	Ongoing or not yet conducted	Report only after validation and analysis

Testing and Verification

The repository contains automated tests for backend models, services, APIs, permissions, geographic processing, Expert System behavior, custom Admin workflows, and mobile functions. These tests constitute implementation evidence when their scope and execution record are known. Historical test files alone are not reported as a current passing result.

During the documentation preparation, all 117 Python source files passed a syntax-tree parse. The full Django and Flutter test suites were not rerun because the available documentation environment lacked Django, Django REST Framework, pytest, psycpg, Flutter, and Dart. Final Chapter 3 test results must identify the environment, date, commit, commands, passed and failed tests, and any unavailable dependencies.

System Evaluation Plan

The final system evaluation will use criteria derived from ISO/IEC 25010:2023 and the approved institutional instrument. The selected characteristics, indicators, scale, respondent groups, administration procedure, and interpretation method will be documented before analysis. Accessibility and context-specific research questions may be reported separately when they are not direct ISO product-quality characteristics.

[AUTHOR NOTE: Survey data collection is ongoing. Complete this section after the validated survey responses and approved analysis are available.]

Statistical Treatment of Data

Frequencies and percentages may be used for completed categorical survey items. When an approved Likert-type evaluation instrument is available, the study may use the institutionally approved scoring and summary method. The manuscript will state the formula, scale anchors, interpretation ranges, number of valid responses, missing-data treatment, and reliability procedure actually used. No weighted mean, ranking, acceptance level, or quality score is reported at this stage.

Ethical, Privacy, and Data-Governance Considerations

Research participation must be voluntary and based on the approved consent and data-handling procedure. Personally identifiable information must be limited to what is necessary, protected from unauthorized access, and retained only according to the approved research process. Application location is requested only when the user initiates the function, used for the active assessment flow, and not continuously collected in the background.

Geographic, susceptibility, rainfall-reference, preparedness, and evacuation-center records must retain source and validation information. Provisional or technically

derived data must be labeled accurately. Operational publication requires the appropriate institutional approval and must not be inferred from repository presence, a diagram, a mockup, or a processing script.

Methodological Limitations

The current prototype demonstrates the application flow and supporting administration but does not yet have an approved operational susceptibility dataset. The 47-barangay boundary is a neutral administrative reference, not a flood-hazard layer. CSV exchange, resident nearest-center discovery, resident account flows, assessment history, notifications, offline operation, final survey analysis, full test execution in the required environments, and production deployment are incomplete or outside the current scope. These limits prevent claims of operational forecasting, official warning capability, or completed user acceptance.

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